Krittin Jindamai (Yim)

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EDUCATION

University of Florida Expected Graduation: May 2027
Bachelor of Science in Aerospace Engineering GPA: 3.98 / 4.00
Davis UWC Scholar | University Scholars Program | Honors Program | Dean's List
Certificate of Outstanding Merit, Herbert Wertheim College of Engineering (2024)

Relevant Coursework: Nonlinear Control | Astrodynamics | Stability and Control of Spacecraft | Optimal Control

RESEARCH EXPERIENCE

Space Systems Group (SSG), Gainesville, Florida
Undergraduate Researcher | Advisor: Dr. Fitz-Coy *Spring 2026 – Present*

Honors Thesis: HCMG Momentum Management Using Magnetorquers

- Built a MATLAB attitude and orbit simulation of a 12U CubeSat for rapid retargeting and precision nadir pointing in sun-synchronous orbit, under gravity-gradient, aerodynamic-drag, and J2 perturbations.
- Implemented generalized singularity-robust steering for the HCMG array, using null motion to reach a nonsingular gimbal configuration before locking into reaction-wheel (RW) mode.
- Applied cross-product magnetorquer desaturation, limiting RW speed growth over 15 simulated orbits; without desaturation, speeds drifted within three orbits, leading to loss of control authority.
- Reviewed passive and active desaturation methods, HCMG steering laws, and spacecraft disturbance models.

HARMONIA: Unified Docking and Debris-Capture Architecture

- Retrieve folded Mini-MegaTORTORA light curves using Python and determine rotation rates for 44 defunct low-Earth-orbit satellites, resolving factor-of-two period ambiguities.
- Estimate grapple radii, the distances from spin axes to capture points, using satellite bus and adapter-ring dimensions, and analyze trends across the dataset.
- Analyzed rotation rate versus time since end of life across the dataset, identifying a trend toward lower rates among satellites inactive for longer.
- Generate Hohmann-phasing and Clohessy–Wiltshire relative-motion plots for far-field rendezvous and proximity operations supporting the lead researcher's PhD dissertation.

Nonlinear Control Research Group (NCR), Gainesville, Florida
Undergraduate Research Assistant | Advisors: Dr. Dixon, Dr. Kamalapurkar *Spring 2026 – Present*

University Scholars Program: CLF-MCBF for Safe HCMG Mode Switching

- Applied matrix-valued control barrier functions to hybrid control moment gyroscope (HCMG) mode switching, solving a constrained semidefinite optimization for safe gimbal locking and unlocking.
- Extending the method to a unified CLF-CBF controller that enforces stability and safety without a nominal controller.
- Validated nonsingular gimbal locking during various maneuvers in MATLAB.
- Currently drafting a conference paper as the first author.

CBF Safety Control for Nonholonomic Robots

- Investigate look-ahead control barrier functions (CBFs) for smooth waypoint tracking of nonholonomic robots near safety constraints.
- Simulated trajectory tracking in a virtual indoor laboratory space using ROS 2 and Gazebo Harmonic to assess feasibility before hardware deployment.

King Mongkut's University of Technology North Bangkok, Thailand
ADCS Research Assistant | Advisor: Dr. Saisutjarit *Summer 2024*

KNACKSAT-3: Attitude Determination and Detumbling

- Implemented TRIAD attitude determination using magnetometer and sun sensor measurements.
- Modeled 1U CubeSat detumbling in low Earth orbit using B-dot control of 3 self-made magnetorquers in MATLAB and Simulink.
- Integrated the nonlinear dynamics using the fourth-order Runge–Kutta method.
- Enabled B-dot controller gain tuning for custom magnetorquers before spacecraft integration.

ENGINEERING PROJECTS

Lunar Hopper Trajectory Optimization and Navigation

Senior Design Project (Astrodynamics and Guidance, Navigation Lead) *Fall 2026*

- Built a minimum-wet-mass outbound-and-return lunar hop optimizer in Python using Hermite–Simpson collocation and IPOPT, enforcing terrain, landing, and sensor-pointing constraints with NASA south-pole elevation data.
- Built GPS-independent navigation with an error-state extended Kalman filter fusing IMU, star tracker, and terrain-relative measurements; achieved RMS landing error below 10 m across 1,000 Monte Carlo simulations with perturbed sensor measurements.
- Informed structural design, propellant-tank sizing, and engine-configuration trades using optimized trajectory and mass results.

Adaptive Attitude Tracking for CubeSats with Uncertain Inertia

Nonlinear Control Final Project *Fall 2026*

- Designed an adaptive attitude-tracking controller for a spacecraft actuated by a four-CMG pyramid with unknown inertia, using a pseudoinverse steering law.
- Proved asymptotic attitude-tracking convergence using Lyapunov analysis and Barbalat's lemma.
- Reduced attitude error from 20° to below 0.001° in about 46 s in a MATLAB simulation, with inertia estimates initialized up to 32% off the true values.

Swamp Launch Rocket Team, Gainesville, Florida

Guidance, Navigation, and Controls Lead Engineer *Fall 2025 – Spring 2026*

- Developed embedded C++ flight software for real-time trajectory prediction and tracking using a PID controller.
- Implemented a Kalman filter to estimate position, quaternion attitude, and IMU biases in real time.
- Designed, assembled, and soldered custom flight-computer PCBs.
- Designed and manufactured a cam-slot airbrake mechanism using CNC machining, milling, and turning.
- Led a team of 11 members to three successful launches within 20 ft of the target altitude.

Payload Software Lead Engineer *Fall 2024 – Spring 2025*

- Led 15 members to a sixth-place overall finish at the 2025 NASA Student Launch Competition.
- Programmed C++ telemetry software to compile and transmit rocket and landing-site conditions to a NASA transceiver over a 2 m-band radio link as an APRS packet.
- Filtered IMU and barometer measurements with a Kalman filter to estimate rocket velocity in real time.
- Collaborated with electrical and mechanical teams to develop the rocket's retention mechanism.

WORK EXPERIENCE

Mechanics of Materials Laboratory, Gainesville, Florida

Teaching Assistant *Fall 2026 – Present*

- Lead two weekly laboratory sections (60+ students) through tensile, compression, and four-point bending tests, guiding data collection with DAQ systems, strain gauges, Wheatstone bridges, and Instron machines.
- Help students connect test results to mechanics of materials theory and follow technical report-writing conventions; hold office hours for courses and lab support.

Housing and Residential Life, Gainesville, Florida

Resident Assistant *Spring 2026 – Present*

- Mediate roommate conflicts and respond to on-call incidents for 34 residents; connect residents with campus support services.
- Organized seven community events and connected residents with campus clubs and design teams.

Geo-Informatics and Space Technology Development Agency (GISTDA), Thailand

Mechanical Engineering Intern *Summer 2025*

- Meshed and assembled the finite-element model of the THEOS-3A primary structure in HyperMesh.
- Characterized structural natural frequencies and mode shapes through free-free and fixed-base modal analyses.
- Evaluated launch and on-orbit structural response through quasi-static analyses at multiple load factors in MSC Nastran.

SKILLS

MATLAB | Simulink | Python | C++ | CasADi | IPOPT | STK | ROS 2 | Gazebo Harmonic | HyperMesh | MSC NASTRAN | Ansys | Fusion | SolidWorks (CSWA Certified) | GitHub | LaTeX